

The Weakening of Aging–Conservatism: Folk Wisdom Meets Korean Politics

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Abstract

This paper revisits the folk wisdom that people become more politically conservative as they age by leveraging the recent South Korean political context. In Korea, individuals born in the 1960s (the 86 generation) were exposed to military dictatorship during their youth, shaping a distinctly liberal political outlook. This historical experience sets up a potential clash between the conventional aging–conservatism link and the persistent liberal orientation of the 86 generation as they enter the elderly population. Using a novel shift–share IV based on regional hospital-type shares and nationwide shifts in elderly medical demand, we find that regional voting patterns become more liberal as the 86 generation crosses the age-55 threshold. Individual-level survey evidence further shows that, unlike other cohorts, the 86 generation maintains a consistently liberal orientation with age. These findings highlight the importance of cohort composition for understanding the political consequences of population aging.

Keywords: Aging, Political Preferences, Shift-Share IV

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1 Introduction

Do people’s political preferences and voting behavior change as they age, or do they remain consistent with their original tendencies? A long-standing piece of folk wisdom holds that individuals become more politically conservative as they age (Peterson et al. 2020). However, preferences and beliefs are inherently abstract and private; they are shaped by complex interactions between individuals and their environments (Druckman and Lupia 2000). This complexity makes it difficult to generalize whether these traits remain stable or evolve over time. While preferences and beliefs may shift in response to certain contexts or shocks (Druckman and Lupia 2016), there is also substantial evidence that they remain remarkably stable (Hanley et al. 2006).

The fields of political economy and political science reflect this same tension. Although a large body of research has examined individuals’ political preferences and beliefs as manifested in voting behavior (Aldrich 1993; Matsusaka and Palda 1999; Grossman and Helpman 2001; Coate and Conlin 2004), the debate between change and persistence remains unresolved. In particular, it is unclear whether aging induces systematic shifts in voting behavior or whether political preferences remain largely fixed once established. On the one hand, several studies support the conventional wisdom that people become more politically conservative as they age (Cornelis et al. 2009; Franssen et al. 2013; Glenn 1974; Kossowska et al. 2012; Tilley and Evans 2014; Van Hiel and Brebels 2011). On the other hand, several empirical studies document considerable stability in political attitudes over time (Achen and Bartels 2017; Campbell et al. 1960; Niemi and Jennings 1991; Peterson et al. 2020; Sears and Funk 1999; Stoker and Jennings 2008). Against this backdrop, where robust causal evidence on the matter is still lacking, this study aims to contribute to the literature by examining the relationship between aging and political outcomes in South Korea, focusing on both regional electoral patterns and individual political orientations.

We focus on the situation in Korea for the following reasons. First, Korea is among the world’s fastest-aging countries. In 2000, individuals aged 15–44 constituted the largest share of the population, while those aged 65 and older accounted for only 7.2%. By 2018, however,

the demographic center had shifted to individuals aged 35–59, and the share of the population aged 65 and older had doubled to 14.3% (Han 2019). According to the 2019 Population Projection by Statistics Korea, the proportion of the population aged 65 and older will reach 37.4%, and by 2060, it is expected to exceed 40%. This demonstrates an unprecedentedly rapid pace of aging on a global scale, making Korea an ideal setting to investigate the effects of population aging.

Second, Korea exhibits a particularly distinctive political divide across age groups. Both the Gallup Korea Daily Opinion survey¹ and the Embrain Trend Monitor² highlight that middle-aged groups, particularly those in their 30s, 40s, and 50s, tend to support the Democratic Party, whereas older individuals are more likely to hold conservative political preferences. This age divide is deeply rooted in Korean history: older generations largely experienced the rapid economic development under the conservative leadership of President Park during the 1960s and 1970s, while the middle-aged groups lived through the student movements and civic activism of the 1980s and 1990s, which fostered strong anti-authoritarian and anti-dictatorship sentiments. This liberal-leaning pattern is especially pronounced among the so-called 86 generation, named for being born in the 1960s and coming of age during the student movements of the 1980s (Noh et al. 2013).

An intriguing aspect of this age divide has emerged as individuals belonging to the 86 generation began entering age 55 and above in the mid-2010s. This development is politically significant because the 86 generation, as part of the baby-boom cohorts, constitutes one of the largest age groups and therefore represents a pivotal electoral bloc. This raises an important question: Will the growing size of the 55-plus population shift the political landscape toward a more liberal orientation if those newly entering this age group retain their original political preferences? Answering this question is crucial, as it is closely connected to the mechanisms underlying the results we aim to uncover, and it ties into existing evidence suggesting that voting behavior reflects the interconnectedness of past experiences

¹<https://www.gallup.co.kr/gallupdb/report.asp>

²<https://trendmonitor.co.kr/tmweb/trend/allTrend/detail.do?bIdx=1689&code=0404&trendType=CKOREA>

and contexts.³ Moreover, investigating this question provides meaningful political implications regarding the consequences of a shifting population distribution, especially as older individuals come to represent a majority of the electorate.

In fact, several sources, including recent polling from Gallup⁴ and the analysis in Noh et al. (2013), provide partial empirical evidence that individuals born between 1960 and 1969 tend to develop liberal voting preferences early in life and persistently maintain these preferences as they age. With richer data and more rigorous methodologies, we aim to extend this discussion by examining whether this phenomenon holds more broadly and whether it ultimately contributes to the weakening of the aging–conservatism link.

To do so, we adopt two complementary empirical strategies. First, we construct a regional panel dataset linking changes in the share of the population aged 55 and older (among the 20-plus population) across Korea’s 229 cities, counties, and districts to recent presidential election outcomes. Specifically, we examine how changes in the 55-plus population share affect (i) the conservative vote share and (ii) the difference between conservative and liberal vote shares. Under the stable-preference narrative, if members of the 86 generation retain their liberal orientation even after crossing age 55, then regions experiencing larger increases in the 55-plus population should exhibit lower conservative vote shares. Furthermore, in historically liberal regions, an increase in the elderly share should raise the winning vote rate and lead to less competitive elections, whereas in historically conservative regions, it should generate more competitive races.

To empirically test these hypotheses while addressing potential endogeneity in demographic shifts, we implement two-stage least squares (2SLS) regressions. Motivated by the idea that population aging naturally increases medical demand among older adults, we adopt a novel shift–share instrumental variable (IV) strategy that leverages regional pre-period hospital-type shares and predicted nationwide shifts in elderly patient demand. By distributing the national-level shift in the share of 55-plus patients for each hospital type across

³Relevant cases include retrospective voting (Healy and Malhotra 2013), cognitive dissonance (Mullainathan and Washington 2009), and responses to industry shocks (Bekhtiar 2025).

⁴see https://www.chosun.com/site/data/html_dir/2020/05/08/2020050800012.html

regions using their pre-period local share of hospitals of that type, we obtain a predicted change in the regional share of individuals aged 55 and older, which serves as a valid IV.

The corresponding results consistently show that increases in the 55-plus population share lead to declines in both the conservative vote share and the conservative–liberal vote share margin. In addition, the heterogeneous effects analysis reveals that the degree of electoral competitiveness differs substantially across historically conservative and liberal regions. Historically liberal regions with larger increases in the 55-plus population tend to experience further gains in liberal electoral strength, whereas historically conservative regions with similar demographic shifts exhibit increasingly competitive races.

After documenting evidence consistent with these predictions using regional variation, we turn to individual-level survey data to examine whether the typical pattern of individuals becoming more conservative with age is indeed attenuated for members of the 86 generation. The analysis reveals that, while most individuals do become more conservative as they age, members of the 86 generation deviate substantially from this pattern, showing a pronounced tendency to maintain their liberal identity. This finding underscores the importance of cohort-specific formative shocks in shaping political trajectories over the life cycle.

The remainder of this paper is organized as follows. Section 2 elaborates on the regional-level analysis including the data, identification strategy using the IV, and the main results. Section 3 discusses the individual-level analysis and its corresponding findings. Finally, Section 4 concludes.

2 Regional-Level Analysis

In this section, we discuss our analysis of leveraging regional variation to examine the causal link between changes in the aging population and corresponding shifts in electoral outcomes. We begin by describing the data, then detail our shift–share instrumental variable strategy, and finally report the estimation results.

2.A Data

We primarily rely on data at the city, county, or district level in Korea. Shares of the 55+ population are constructed from the Year-Centered Census Data of the Population and Housing Census provided by Statistics Korea. As shown in the relevant summary statistics in Table 1, the average share of the 55+ population increases steadily from 2012 to 2025. This upward trend also holds when we scale the 55+ population by the 20+ population.⁵

The primary data source for election outcomes is the Election Statistics System on the National Election Commission website. This system covers recent elections, including the 2017, 2022, and 2025 presidential elections. For each presidential election, it provides city, county, or district level data on the number of eligible voters, total votes cast, votes by candidate (or party), invalid votes, and abstentions. Summary statistics for the three most recent presidential elections are reported in Table 2. Here, the conservative vote share is the share of votes received by candidates from conservative-leaning parties, and the vote share margin is the difference between conservative and liberal vote shares. The standard deviations and the min–max ranges indicate substantial cross-regional variation and pronounced polarization.

Data for control variables were compiled from multiple sources: the Population and Housing Census (Statistics Korea), the Real Transaction Price Disclosure System (Ministry of Land, Infrastructure, and Transport), Status of Urban Planning (Korea Land and Geospatial Informatix Corporation), Domestic Migration Statistics (Statistics Korea), local GRDP statistics (local governments), and the Survey on Labor Conditions by Establishment (Ministry of Employment and Labor). Data for the instrumental variable come from the National Health Insurance Statistics and the Healthcare Bigdata Hub provided by the Health Insurance Review and Assessment Service. Section 2.B provides a detailed description of these variables.

⁵Because Korea lowered the legal voting age in 2019 (from 19+ to 18+), we scale the 55+ population by the 20+ population in the main analysis to keep the population base consistent over time. However, using the voting-eligible population as the denominator yields very similar result patterns.

Table 1: Descriptive Statistics for the Share of Population Aged 55 and Older

	number of obs.	mean	std. dev.	min.	max
Among the total population					
55+ share in 2012	229	0.289	0.096	0.126	0.505
55+ share in 2017	229	0.350	0.101	0.176	0.587
55+ share in 2022	229	0.412	0.112	0.211	0.664
55+ share in 2025	229	0.451	0.115	0.237	0.715
Among the 20+ population					
55+ share in 2012	229	0.358	0.104	0.178	0.576
55+ share in 2017	229	0.420	0.106	0.239	0.649
55+ share in 2022	229	0.478	0.114	0.272	0.710
55+ share in 2025	229	0.516	0.116	0.302	0.758

Notes: The units of observation are regions (city/county/district).

Table 2: Descriptive Statistics for Conservative Vote Shares

	number of obs.	mean	std. dev.	min.	max
Conservative Vote Share					
2017	229	0.332	0.175	0.036	0.728
2022	229	0.487	0.196	0.089	0.834
2025	229	0.496	0.193	0.093	0.847
Conservative-Liberal Vote Share Margin					
2017	229	-0.118	0.299	-0.674	0.572
2022	229	-0.015	0.392	-0.812	0.681
2025	229	-0.007	0.386	-0.811	0.694

Notes: "Conservative Vote Share" is the share of votes received by candidates from conservative-leaning parties.
 "Conservative-Liberal Vote Share Margin" is the difference between conservative and liberal vote shares.

2.B Empirical Framework

To examine the relationship between aging and political preferences, it is natural to consider simple regressions that exploit regional variation in changes in the share of residents aged 55 and over and in electoral outcomes. However, two major endogeneity concerns remain. First, regional dynamics of the elderly population are unlikely to be random: changes in the 55+ share may be driven by unobserved and uncontrolled, time-varying regional shocks. For example, an unobservable amenity shock could simultaneously attract older residents and shift preferences toward (or away from) certain responsible political parties, biasing the estimates. Second, reverse causality may arise if a region’s age structure is itself affected by political events related to elections. To address these issues, we employ a shift-share (or Bartik-style) instrumental variable strategy, detailed in Section 2.B.1.

Using this instrument, we estimate two-stage least squares (2SLS) models to identify the causal effects of changes in the 55+ population share on the conservative vote share and on the conservative–liberal vote-share margin. We then examine heterogeneous effects on electoral competitiveness by historical regional partisan orientation. In all 2SLS specifications, we control for a rich set of regional characteristics that could potentially bias the estimates, incorporate region and time (window) fixed effects to absorb unobserved confounders, weight observations by electorate size in the initial year of the window, and cluster standard errors at the region level.

2.B.1 Econometric Specifications and Instrumental Variable Approach

$$Y_{rt} = \beta_0 + \beta_1 \Delta 55^+ share_{rt} + \mathbf{X}_{rt-1}' \beta_2 + \gamma_r + \delta_t + \epsilon_{rt} \quad (1)$$

Above is the baseline empirical specification pertaining to the regional-level OLS regression. The dependent variable, Y_{rt} , represents the electoral outcomes in region (city, country, or district) r in election period $t \in \{2017, 2022, 2025\}$. These outcomes include conservative vote share and conservative-liberal vote share margin.⁶ The main independent variable

⁶Results for turnout are reported in the Appendix.

, $\Delta 55^+ share_{rt}$, is the change in the share of the 55-plus population relative to the 20-plus population from election period $t - 1$ to t .

$\mathbf{X}_{rt-1}'$ is the vector of regional-level controls measured in the initial year of each election window ($t - 1$). It consists of various regional traits reflecting age demographics (share of individuals in their 20s, 30s, 40s among the 20-plus population)⁷, gender composition (male share among the 20-plus population), educational attainment (shares of high, medium, and low education groups, with “low” as the omitted category)⁸, regional economic conditions (logged GRDP per capita, employment-to-population ratio, and value-added and employment shares by sector, with the primary sector as the baseline), degree of urbanization (share of the population living in urbanized areas), real estate market prices (logged prices for housing, buildings, and land), and migration (net migration share). In some specifications, we additionally control for changes in the shares of incoming and outgoing migrants aged 55-plus among the total 55-plus population from $t - 1$ to t to account for time-varying demographic shifts driven by migration.

Importantly, as the data is structured as a panel, incorporating two-way fixed effects into the specification becomes advantageous. This allows for comparisons within each region and within each time period. In particular, as political preferences are strongly correlated with regional identity and geography, the region fixed effects, denoted as γ_r , purge unobserved and time-invariant regional factors that might otherwise bias voting outcomes.⁹ Additionally, the time fixed effects, denoted as δ_t , help screen out the effects of nationwide political contexts and period-specific shocks.¹⁰

The coefficient of interest is β_1 , which captures the estimated relationship between the percentage-point change in the 55-plus population share and the percentage-point change in

⁷Since the dependent variable reflects total electoral outcomes rather than age-specific outcomes, controlling for these variables helps filter out the effects of other demographic shifts and isolate the impact caused by change in the share of 55-plus population.

⁸Education groups are defined by highest attainment: graduate degree (high), undergraduate degree (medium), and high-school or below (low).

⁹For example, regions in the Jeolla Province area typically lean liberal, while regions in the Gyeongsang Province area tend to have a conservative stance.

¹⁰For example, the 2016 presidential impeachment and the Justice Minister Cho scandal in 2019 had significant ramifications in Korean politics.

electoral outcomes. However, identifying a causal effect requires careful attention to potential endogeneity. Despite the inclusion of a wide array of observable controls in the proposed specification, unobservable omitted variables that are correlated with both population aging and voting trends may still exist.

To mitigate these endogeneity concerns that may lead to estimation biases, we adopt a novel shift-share (or Bartik-style) IV technique. This approach leverages regional variation in the lagged number of hospitals by type, interacted with the national-level contemporary change in the predicted share of the 55-plus population seeking care from each hospital type, scaled by the 20-plus population.¹¹

$$SSIV_{rt} = \sum_h s_{rht-2} \Delta_{ht}, \quad (2)$$

where

$$s_{rht-2} = \left(\frac{\# \text{ type } h \text{ hospital}}{\text{total } \# \text{ hospitals}} \right)_{r,t-2} \text{ and } \Delta_{ht} = \Delta \left(\frac{\widehat{55^+ \text{ patients}_h}}{20^+ \text{ population}} \right)_{\text{nation},t}$$

Specifically, the share component, denoted as s_{rht-2} , refers to the baseline proportion of type h hospitals among all hospitals in region i at period $t - 2$, which corresponds to six years prior to the election period $t - 1$. The set of hospital types h includes family medicine, internal medicine, urology, obstetrics and gynecology, plastic surgery, neurology and neurosurgery, ophthalmology, general surgery, otorhinolaryngology (ENT), psychiatry, orthopedics/rehabilitation/anesthesiology & pain medicine, dermatology, dentistry, traditional Korean medicine hospitals, and traditional Korean medicine clinics.¹²

The shift component, denoted as Δ_{ht} , captures the national-level change from period

¹¹Relevant data is available from National Health Insurance Statistics and Healthcare Bigdata Hub from Health Insurance Review & Assessment Service.

¹²For hospitals classified as specialists without designated major fields and general practitioners, whose primary treatment domain is not explicitly identified, we allocate these hospitals to one of the listed types based on statistics from the Health Insurance Review and Assessment Service that report the proportions of such hospitals affiliated with each specialty category. In regions where the total number of hospitals is fewer than 15, we assign these hospitals to family medicine, reflecting the fact that family medicine commonly provides most broad and general medical care across a wide range of diseases and patients.

$t - 1$ to t in the share of patients aged 55 or over who access medical facilities for disease categories primarily treated by type h hospitals, relative to 20-plus population. This variable is constructed using the following steps:

1. Using the information on the number of patients by age group and by 298 disease categories from the National Health Insurance Statistics, obtain the number of individuals aged 55 and over diagnosed with each disease category.
2. Classify each disease category into its most relevant hospital type using the 298 Disease Classification Table and the Korean Standard Classification of Diseases (KCD) code.
3. Combine these two sources of information to predict the number of individuals aged 55 and over who visited each type of hospital.

Examples of the disease-category–hospital-type mapping are provided in Table 3. For illustration, consider a hypothetical scenario in which the national number of elderly individuals diagnosed with H40–H42 (glaucoma) is 200 and the number diagnosed with M15–M19 (arthritis) is 300. Following the procedure described above, the algorithm would assign 200 to ophthalmology and 300 to orthopedics.¹³

The rationale behind the relevance condition of the IV lies in the inferred ability of the combination of the national-level shift and the region-level share to predict the independent variable. As population aging is accompanied by increasing health demand among older adults, national shifts in the share of 55-plus patients for relevant hospital type h would be sizable. Given the shift-share IV structure, regions with higher baseline exposure s_{rht-2} load more heavily on these national shifts. Thus, if older patients nationwide increasingly seek treatment for diseases primarily treated by hospital type h , regions with greater specialization in h would more likely to experience larger retention or inflows of older residents. In other

¹³Due to data limitations, it is not possible to determine the number of individuals simultaneously diagnosed with multiple diseases (e.g., both glaucoma and arthritis). This limitation may raise concerns regarding the relevance condition of the instrument. However, we view this as a minor issue, as first-stage F-statistics reported in Table 4 and Table 6 are sufficiently large.

Table 3: Disease-Hospital Mapping Examples

Disease Name	KCD Code	Hospital Type
Tuberculosis (결핵)	A15–A19	Internal Medicine (내과)
Diabetes Mellitus (당뇨병)	E10–E14	Family Medicine (가정의학과)
Mood Disorders (기분 장애)	F30–F39	Psychiatry (정신건강의학과)
Glaucoma (녹내장)	H40–H42	Ophthalmology (안과)
Varicose Veins (하지정맥류)	I83	General Surgery (외과)
Pharyngitis/Tonsillitis (인두염 및 편도염)	J02–J03	ENT (이비인후과)
Skin Infections (피부 감염)	L00–L08	Dermatology (피부과)
Arthritis (관절증)	M15–M19	Orthopedics (정형외과)
BPH (전립선 비대증)	N40	Urology (비뇨의학과)

Notes: The 298-category disease list does not comprehensively cover conditions treated in Dentistry, Korean Medicine Hospital, and Korean Medicine Clinic. For these three types, I supplement with patient-volume data from the Healthcare Bigdata Hub.

words, following the standard logic and contemporary interpretation embedded in shift-share IVs (Goldsmith-Pinkham et al. 2020; Borusyak et al. 2021), the national-level shift value would be distributed across regions proportionally to each region’s pre-determined hospital-type shares. Under this structure, information that older people were more frequently diagnosed with particular diseases at the national level would translate into the prediction that these elderly individuals would be more concentrated in regions with a higher share of hospitals specializing in the treatment of those specific diseases.

Crucially, the designed shift-share IV is designed to follow the exogenous share framework emphasized by Goldsmith-Pinkham et al. (2020). Thus, the underlying fundamental assumption regarding the exclusion restriction condition is that, conditional on control variables, pre-measured hospital type shares are uncorrelated with unobservable determinants of future electoral outcomes. Because these baseline shares are measured well before the sample window and even before the main analysis window, and even before the election held prior to the election at period $t - 1$, it is unlikely that the identifying share variation directly correlates to the dependent variables of interest. Moreover, the temporal structure of the instrument design minimizes the possibility that contemporaneous political dynamics influence the instrument. Indirect empirical assessments of the exclusion restriction are

provided in Section 2.B.2.

Given the mainline empirical strategy, we further aim to examine how the concentration of vote shares varies heterogeneously across historically conservative and liberal regions. To do so, we estimate the following similar specification:

$$Y_{rt} = \beta_0 + \beta_1 \Delta 55^+ share_{rt} + \theta (\Delta 55^+ share_{rt} * C_r) + \mathbf{X}_{rt-1}' \beta_2 + \gamma_r + \delta_t + \epsilon_{rt}, \quad (3)$$

where C_r is an indicator equal to 1 if the conservative party won at least two of the 1997, 2002, 2007, and 2012 presidential elections in region r . For the dependent variable, we consider the winning vote share and the Blau Index.¹⁴ These variables are defined as follows:

$$\text{Winning Vote Share} = \left(\frac{\text{winning vote count}}{\text{valid vote count}} \right)_{rt},$$

$$\text{Blau Index} = 1 - \sum_{j=1}^{n_c} p_j^2,$$

where n_c is the number of candidates and p_j is the vote share (probability) that candidate j received (Gini 1912; Simpson 1949; Nunes et al. 2020). Following this construction, lower values of the index indicate increasingly one-sided races, whereas higher values indicate greater electoral competition. As there are two endogenous variables in this specification, $\Delta 55^+ share_{rt}$ and $\Delta 55^+ share_{rt} * C_r$, we instrument them using both $SSIV_{rt}$ and its interaction $SSIV_{rt} * C_r$.

2.B.2 IV First-Stage and Falsification Tests

Columns (1) and (2) of Table 4 present the formal first-stage regression results. The explanatory variable is the shift-share IV, and the dependent variable is the change in the 55-plus population share. Across both columns, the IV coefficients are positive and statistically significant, regardless of whether the migration shares change controls is included.¹⁵

To assess whether the logic of the IV construction is well reflected in the empirical re-

¹⁴The Blau Index is also known as the Gini-Simpson Index.

¹⁵The corresponding first-stage F-statistics are reported in Table 6, confirming the instrument's relevance.

sults, we conduct a set of placebo tests examining whether the instrument predicts changes in the shares of younger age groups.¹⁶ Since the shift component is based on the number of patients aged 55 and over, the IV should not be significantly correlated with changes in the shares of unrelated younger groups.¹⁷ Columns (3) through (8) of Table 4 report the results. As expected, none of the estimates in these columns are statistically significant, providing supportive evidence for the validity of the IV.

Table 4: IV First Stage and Placebo Tests

Dependent Variable:	$\Delta 55^+ share_{rt}$		$\Delta 0 - 19 share_{rt}$		$\Delta 20s share_{it}$		$\Delta 30s share_{it}$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SSIV_{rt}$	0.665*** (0.125)	0.703*** (0.125)	0.053 (0.079)	0.053 (0.079)	-0.107 (0.102)	-0.104 (0.104)	0.085 (0.125)	0.047 (0.140)
Controls (pre-period)	✓	✓	✓	✓	✓	✓	✓	✓
$\Delta 55^+$ Migrant Shares		✓		✓		✓		✓
$\Delta 55^+ share_{rt}$			✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓
R^2_{adj}	0.845	0.853	0.880	0.883	0.629	0.637	0.753	0.754
Number of Regions	229	229	229	229	229	229	229	229
Observations	687	687	687	687	687	687	687	687

*p < 0.10, **p < 0.05, ***p < 0.01.

The identifying assumption underlying the IV strategy is that, conditional on the control variables, the instrument affects electoral outcomes solely through its impact on the change in share of people aged 55 and over. Although this exclusion restriction cannot be tested directly, we assess the instrument's validity indirectly by examining whether the IV is significantly correlated with (i) contemporary election outcomes (reduced-form analysis) for the

¹⁶I omit age demographics controls (shares of specific age groups) from these placebo regressions because they partially overlap with the dependent variables.

¹⁷Because a large increase in the 55-plus share could mechanically influence the shares of other age groups, as the total must sum to one, I control for $\Delta 55^+ share_{rt}$ in all placebo regressions.

periods $t = 2017, 2022, 2025$, and (ii) pre-period election outcomes (pre-trend falsification tests) for the periods $t - 3 = 2002, 2007, 2012$. The corresponding regression specifications are:

$$Y_{rt} = \alpha_0 + \alpha_1 SSIV_{rt} (+ \mathbf{X}_{rt-1}' \alpha_2) + \gamma_r + \delta_t + u_{rt} \quad (4)$$

$$Y_{rt-3} = \alpha_0 + \alpha_1 SSIV_{rt} (+ \mathbf{X}_{rt-1}' \alpha_2) + \gamma_r + \delta_t + u_{rt-3} \quad (5)$$

As reported in Table 5, the reduced-form coefficients in columns (1) through (4) are statistically significant, whereas the pre-trend estimates in columns (5) through (8) are smaller in magnitude and statistically insignificant. This pattern supports the plausibility of the exclusion restriction: the IV predicts contemporary political outcomes but is not spuriously correlated with pre-existing electoral trends.

Table 5: IV Reduced Form and Pre-Trend Tests

Dependent Variable:	Conservative Vote Share _{rt}		C-L Vote Margin Share _{rt}		Conservative Vote Share _{rt-3}		C-L Vote Margin Share _{rt-3}	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>SSIV_{rt}</i>	-1.338*** (0.307)	-1.317*** (0.335)	-2.555*** (0.694)	-2.903*** (0.809)	0.083 (0.440)	0.121 (0.555)	-0.563 (1.126)	-0.644 (1.297)
Controls (pre-period)		✓		✓		✓		✓
Region FE	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓
R^2_{adj}	0.976	0.984	0.955	0.965	0.945	0.954	0.911	0.924
Number of Regions	229	229	229	229	229	229	229	229
Observations	687	687	687	687	687	687	687	687

*p < 0.10, **p < 0.05, ***p < 0.01.

2.C 2SLS Results

2.C.1 Impact on Conservative Vote Share

The 2SLS results capturing the causal effects of aging on conservative vote shares and the conservative–liberal vote share margin are presented in Table 6. Across columns (1) through

(3), which incorporate different combinations of control variables, the results consistently show a statistically significant decline in conservative vote share. In particular, the estimate in column (3), which includes sets of pre-period controls as well as the migration shares change controls indicates that a 1 percentage-point increase in the 55-plus population share leads to roughly a 1.5 percentage-point reduction in the conservative vote share.

The results for the conservative–liberal vote share margin, reported in columns (4) through (6), reveal a similar pattern. The larger magnitudes of these coefficients (e.g., a 3.5 percentage-point decline in column (6)) suggest that the loss in conservative vote share is accompanied by a corresponding gain in liberal vote share, thereby widening the gap between the two sides.

Table 6: Effect on Conservative Vote Shares - 2SLS

Dependent Variable:	Conservative Vote Share _{rt}			C-L Vote Margin Share _{rt}		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta 55^+ share_{rt}$	-2.329*** (0.710)	-1.821*** (0.510)	-1.556*** (0.461)	-4.446*** (1.481)	-4.015*** (1.226)	-3.564*** (1.161)
Region FE	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓
Controls (pre-period)		✓	✓		✓	✓
$\Delta 55^+$ Migrant Shares			✓			✓
1st-stage Estimate ($SSIV_{it}$)	0.575*** (0.134)	0.723*** (0.110)	0.735*** (0.109)	0.575*** (0.134)	0.723*** (0.110)	0.735*** (0.109)
1st-stage F-stat (Kleibergen-Paap rk Wald)	18.43	43.07	45.28	18.43	43.07	45.28
1st-stage F-stat (Montiel-Pflueger)	12.26	28.12	29.52	12.26	28.12	29.52
Number of Regions	229	229	229	229	229	229
Observations	687	687	687	687	687	687

*p < 0.10, **p < 0.05, ***p < 0.01.

2.C.2 Heterogeneous Effects on Electoral Competitiveness

Table 7 reports the estimation results pertaining to specification (3), which examines the heterogeneous effects of aging on electoral competitiveness depending on a region's historical political orientation. The results show that in historically liberal regions, increases in the share of residents aged 55 and over are associated with higher winning vote shares. In historically conservative regions, however, this effect is smaller, and the results in column (3) suggest that the winning vote share actually declines: the sum of the main effect and the interaction term is negative.

The Blau Index results exhibit a similar pattern. In liberal regions, the index moves toward 0, indicating increasingly one-sided races. In conservative regions, by contrast, the sum of the relevant coefficients is positive, implying that aging leads to more competitive races.

Table 7: Heterogeneous Effects on Electoral Competitiveness - 2SLS

Dependent Variable:	Winning Vote Share _{rt}			Blau Index _{rt}		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta 55^+ share_{rt}$	2.100** (0.865)	3.211*** (0.915)	2.789*** (0.846)	-1.705*** (0.585)	-2.495*** (0.619)	-2.130*** (0.573)
$\Delta 55^+ share_{rt} * C_r$	-3.205*** (0.445)	-2.968*** (0.603)	-2.885*** (0.561)	3.333*** (0.380)	3.222*** (0.479)	3.157*** (0.429)
Region FE	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓
Controls (pre-period)		✓	✓		✓	✓
$\Delta 55^+$ Migrant Shares			✓			✓
1st-stage F-stat (Kleibergen-Paap rk Wald)	6.76	20.28	20.54	6.76	20.28	20.54
Number of Regions	229	229	229	229	229	229
Observations	687	687	687	687	687	687

*p < 0.10, **p < 0.05, ***p < 0.01.

2.C.3 Further Analysis and Robustness Checks

The 2SLS results presented thus far support the main narrative: the persistence of the original liberal orientation of the core 86 generation, who have recently crossed the age-55 threshold and constitute a substantial share of the increase in the elderly population, causally contributes to the observed link between aging and liberal electoral strength. Consequently, historically liberal regions experiencing a larger inflow of the 86 generation into the 55-plus population tend to see liberal candidates winning by even wider margins. In contrast, historically conservative regions with a similarly large inflow of the 86 generation observe more competitive races between conservative and liberal parties.

One potential concern is that short-term fluctuations in the elderly population, rather than the inflow of the 86 generation, may be driving the variation in $\Delta 55^+ share_{rt}$. To address this, we conduct a robustness check using the change in the population share of individuals aged 55–64 as an alternative independent variable. The results continue to show significantly negative effects, with larger magnitudes. Results are available upon request.

Another concern is that outlier regions with extremely one-sided and relatively unchanging political preferences (usually regions in Gyeongsang and Jeolla Province) may disproportionately influence the estimates, potentially weakening the proposed mechanism. To address this possibility, we replicate the main 2SLS analysis after excluding the top and bottom 5 percent of regions based on their average conservative vote share in the 1997, 2002, 2007, and 2012 presidential elections. Reassuringly, the results remain consistently negative and statistically significant. Results are available upon request.

2.C.4 IV Diagnostics

As formalized by [Goldsmith-Pinkham et al. \(2020\)](#), the Bartik IV estimator $\hat{\beta}^{\text{Bartik}}$ can be expressed as a weighted average of the estimates that would be obtained if each interaction of the share and shift for a single h type were used as the sole instrument, as shown in equation (6). The corresponding weights $\hat{\alpha}_h$, known as Rotemberg weights, indicate the relative contribution of each h type to the identification of the Bartik IV estimate. A larger weight

implies that the corresponding h type is more influential for identification.

Because our analysis uses panel data with multiple time periods, we follow the aggregation procedure suggested by Goldsmith-Pinkham et al. (2020), that aggregates the h, t -level Rotemberg weights into h -level quantities, as shown in equation 7.

$$\hat{\beta}^{\text{Bartik}} = \sum_h \hat{\alpha}_h \hat{\beta}_h, \quad (6)$$

$$\hat{\alpha}_h \equiv \sum_t \hat{\alpha}_{h,t}, \quad \hat{\beta}_h \equiv \sum_t \left(\frac{\hat{\alpha}_{h,t}}{\hat{\alpha}_h} \right) \hat{\beta}_{h,t} \quad (7)$$

Table 8 and Table 9 present valuable information related to Rotemberg weights computed for the specification in which the dependent variable is the conservative vote share and all baseline controls are included (corresponding to the estimation result reported in column (3) of Table 6).¹⁸

Table 8 separates the negative and positive Rotemberg weights and reports, for each category, the sum, mean, and share of the total weight. As shown, the vast majority of the weights are positive, mitigating potential concerns associated with negative Rotemberg weights.

Table 8: Negative and Positive Rotemberg Weights

	Sum	Mean	Share	$\sum_h \alpha_h \beta_h$
Negative RW	-0.104	-0.017	0.086	-0.243
Positive RW	1.104	0.123	0.914	-1.291

Table 9 reports the three hospital types with the largest Rotemberg weights. The results indicate that Family Medicine, Dentistry, and Orthopedics receive the highest weight values, implying that the exogeneity of the pre-period shares of these hospital types is especially important for identification.

Thus, following the logic of the pre-trend specification (5), we examine whether the baseline shares measured at $t - 2 = 2006, 2011, 2016$ for Family Medicine, Dentistry, and

¹⁸Similar results are obtained when Rotemberg weights are computed for the specification in which the dependent variable is the conservative–liberal vote share margin.

Table 9: Top Three Rotemberg Weight Hospital Types

Hospital Type	α_h	β_h
Family Medicine (가정 의학과)	0.677	-0.832
Dentistry (치과)	0.220	-1.086
Orthopedics (정형외과)	0.095	-1.568

Orthopedics are significantly correlated with the pre-period ($t - 3 = 2002, 2007, 2012$) conservative vote share and vote share margin by estimating specification (8). Reassuringly, all of the coefficient estimates reported in Table 10 are statistically insignificant.

$$Y_{rt-3} = \alpha_0 + \alpha_1 \text{Share}_{rt-2}^{(\text{hospital type of interest})} (+ \mathbf{X}_{rt-1}' \alpha_2) + \gamma_r + \delta_t + u_{rt-3} \quad (8)$$

Table 10: Additional Pre-Trend Tests

Dependent Variable:	Conservative Vote Share $_{rt-3}$			C-L Vote Margin Share $_{rt-3}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Share $_{rt-2}^{(\text{Family Medicine})}$	-0.224 (0.139)			-0.381 (0.302)		
Share $_{rt-2}^{(\text{Dentistry})}$		0.117 (0.135)			0.305 (0.293)	
Share $_{rt-2}^{(\text{Orthopedics})}$			0.222 (0.309)			0.315 (0.678)
Controls (pre-period)	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓
Number of Regions	229	229	229	229	229	229
Observations	687	687	687	687	687	687

3 Individual-Level Analysis

In this section, we discuss our analysis of leveraging individual-level variation to more directly investigate whether the proposed mechanism involving the 86 generation holds true. We begin by describing the data, then outline the setup of two individual-level regressions used in the analysis, and finally present the estimation results.

3.A Data

The individual-level data come from the Korean General Social Survey (KGSS), administered by the Sungkyunkwan University Survey Research Center. The KGSS is a repeated cross-sectional survey collected annually from 2003 to 2014 and biannually from 2014 to 2023, and contains information on respondents' ages, political orientations, and a wide range of individual characteristics.¹⁹ The main political orientation (identity) variable is measured on a Likert scale ranging from 1 (“very liberal”) to 5 (“very conservative”).

Table 11: Descriptive Statistics for Key Individual Characteristics

	# obs.	mean	std. dev.	min.	max
Political Identity	21,068	3.016	0.977	1	5
Age	22,050	46.12	16.86	18	99
86 Generation (Dummy)	22,050	0.235	0.424	0	1
Gender (Dummy)	22,071	0.457	0.498	0	1
Employed (Dummy)	22,070	0.580	0.494	0	1
Religion (Dummy)	22,017	0.571	0.495	0	1
Years of Education	22,012	12.41	4.269	0	23
Respondent's Monthly Income	14,343	185.0	283.7	0	10,000
Household Monthly Income	16,912	398.6	510.6	0	33,000

¹⁹Due to COVID-19, the survey collected data in 2021 instead of 2020.

Table 11 reports summary statistics for the data. As shown in the table, the mean political orientation is close to 3, indicating that the sample is not skewed toward either ideological extreme. Likewise, age, gender, religion, employment status, years of education, income, and other key demographic variables are also reasonably well balanced, as suggested by their means and standard deviations. Importantly, individuals belonging to the 86 generation constitute roughly 25 percent of the full sample, ensuring sufficient coverage for our analysis.

3.B Individual-Level Regressions

We first estimate the following regression, which is designed to examine how political orientations evolve once individuals reach age 50, depending on whether they belong to the 86 generation (those born in the 1960s).²⁰

$$PI_{iy} = \alpha \text{Age}_{iy} + \beta (\mathbb{I}[\text{Age}_{iy} \geq 50]) + \theta (\mathbb{I}[\text{Age}_{iy} \geq 50] * \mathbb{I}[c(i) \in 1960s]) + \gamma' \mathbf{X}_{iy} + \lambda_{ry} + \kappa_{g(i)} + \varepsilon_{iy} \quad (9)$$

Here, i denotes an individual and y denotes the survey year. The dependent variable PI_{iy} represents the political identity of individual i in year y . The indicator $\mathbb{I}[c(i) \in 1960s]$ equals 1 if individual i belongs to the 86 generation. As shown on the right-hand side of the specification, we include controls for a variety of individual characteristics ($\mathbf{X}_{iy}$) and region-by-year fixed effects (λ_{ry}). We also include fixed effects for the generation group to which each respondent belongs, denoted as $g(i)$.²¹ Standard errors are clustered at the year-by-generation-group level.

The corresponding results are presented in Table 12 below. Across various combinations of control variables and fixed effects, we consistently find that individuals who do not belong

²⁰We set the age cutoff at 50 rather than 55 because, given the characteristics of the sample, the number of 86-generation respondents aged 55 or older is relatively small compared to those below 55. Using 50 as the cutoff helps maintain better sample balance. Nevertheless, when we re-estimate the model using 55 as the cutoff, the overall pattern of results remains very similar, although the statistical significance becomes slightly weaker. Results are available upon request.

²¹The generation groups are classified as follows: pre-colony (born before 1945), post-colony and pre-baby boom (born 1946–1954), baby boom (born 1955–1963), Generation X and Generation 86 (born 1964–1969), Generation X but not Generation 86 (born 1970–1979), Generation M (born 1980–1994), and Generation Z (born after 1995).

to the 86 generation exhibit a statistically significant shift toward conservatism once they pass age 50. In contrast, members of the 86 generation display the opposite pattern, showing a tendency to remain, if not become even more, aligned with the liberal side.

Table 12: Individual-Level Regression 1

Dependent variable: PI_{iy}						
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{I}[\text{Age}_{iy} \geq 50]$	0.104** (0.046)	0.105** (0.046)	0.100** (0.047)	0.100** (0.048)	0.102** (0.049)	0.117* (0.062)
$\mathbb{I}[\text{Age}_{iy} \geq 50] * \mathbb{I}[c(i) \in 1960s]$	-0.136** (0.061)	-0.136** (0.061)	-0.139** (0.060)	-0.140** (0.060)	-0.142** (0.061)	-0.157** (0.064)
age	✓	✓	✓	✓	✓	✓
generation group FE	✓	✓	✓	✓	✓	✓
region-by-year FE	✓	✓	✓	✓	✓	✓
gender		✓	✓	✓	✓	✓
education			✓	✓	✓	✓
marriage status				✓	✓	✓
religion					✓	✓
industry FE and labor-related						✓
Observations	20,966	20,966	17,850	17,842	17,808	11,125

*p < 0.10, **p < 0.05, ***p < 0.01.

Because age 50 is an arbitrary cutoff, we also estimate the following regression that does not impose a specific threshold. Instead, we interact the continuous age variable with a dummy for being born in the 1960s, with the remaining components following a setup similar to the previous specification:

$$PI_{iy} = \beta \text{Age}_{iy} + \theta (\text{Age}_{iy} * \mathbb{I}[c(i) \in 1960s]) + \gamma' \mathbf{X}_{iy} + \lambda_{ry} + \kappa_{g(i)} + \varepsilon_{iy} \quad (10)$$

The corresponding results are presented in Table 13. Consistent with the previous findings, individuals generally become more conservative as they age; however, this pattern

weakens substantially for members of the 86 generation.

Table 13: Individual-Level Regressions

Dependent variable: Political Identity _{iy}						
	(1)	(2)	(3)	(4)	(5)	(6)
Age _{iy}	0.011*** (0.003)	0.011*** (0.003)	0.011*** (0.003)	0.011*** (0.003)	0.011*** (0.003)	0.011*** (0.004)
Age _{iy} * $\mathbb{I}[c(i) \in 1960s]$	-0.003*** (0.001)	-0.003*** (0.001)	-0.003** (0.001)	-0.003** (0.001)	-0.003** (0.001)	-0.003** (0.001)
age	✓	✓	✓	✓	✓	✓
generation group FE	✓	✓	✓	✓	✓	✓
region-by-year FE	✓	✓	✓	✓	✓	✓
gender		✓	✓	✓	✓	✓
education			✓	✓	✓	✓
marriage status				✓	✓	✓
religion					✓	✓
industry FE and labor-related						✓
Observations	20,966	20,966	17,850	17,842	17,808	11,125

*p < 0.10, **p < 0.05, ***p < 0.01.

4 Conclusion

This study explores the “aging–conservatism” narrative by examining how recent population aging shaped regional political outcomes in Korea, with particular attention to the 86 generation—a cohort that experienced distinctive formative political and social shocks during its youth. Using a novel shift–share IV strategy that leverages regional pre-period hospital-type shares and predicted nationwide shifts in elderly patient demand, we find consistent evidence that the 86 generation’s persistent liberal orientation exerts a strong influence on regional voting patterns as this cohort crosses the age-55 threshold. Consequently, historically liberal regions with greater inflows of the 86 generation into the 55-plus population tend

to experience further gains in liberal electoral strength, whereas historically conservative regions with similar demographic shifts exhibit increasingly competitive races. Complementary individual-level analyses reinforce this interpretation: while most individuals become more conservative with age, members of the 86 generation deviate substantially from this pattern, showing a pronounced tendency to maintain their liberal identity. This highlights the importance of cohort-specific formative shocks in shaping political trajectories over the life cycle.

Taken together, these results underscore that the widely held folk wisdom that “people naturally become conservative as they age” does not uniformly apply. Instead, the political impact of demographic aging depends critically on which cohorts are entering old age and on the historical experiences that formed their political identities. As many other countries underwent rapid demographic transitions, understanding the political consequences of aging in these societies would require careful attention to cohort composition rather than relying on simple generalizations linking age and ideology.

One important limitation of our analysis is that it does not directly address potential heterogeneity between ideological orientation and actual voting behavior. For instance, individuals may become more conservative in their attitudes as they age while continuing to vote for liberal parties due to habit or long-standing partisan attachments. Investigating these potential divergences between political identity, issue preferences, and voting behavior represents an important direction for future research.

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Appendix

A Impact on Turnout

Table A1: Effect on Turnout - OLS and 2SLS

Dependent variable: Turnout _{rt}						
	(1)	OLS (2)	(3)	(4)	2SLS (5)	(6)
$\Delta 55^+ share_{rt}$	0.028 (0.038)	-0.015 (0.058)	0.019 (0.061)	1.113*** (0.291)	0.826*** (0.248)	0.818*** (0.239)
Region FE	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓
Controls (pre-period)		✓	✓		✓	✓
$\Delta 55^+$ Migrant Shares			✓			✓
R^2_{adj}	0.884	0.906	0.907			
1st-stage Estimate ($SSIV_{it}$)				0.575*** (0.134)	0.723*** (0.110)	0.735*** (0.109)
1st-stage F-stat				18.44	43.07	45.28
Number of Regions	229	229	229	229	229	229
Observations	687	687	687	687	687	687

*p < 0.10, **p < 0.05, ***p < 0.01.

B More details on Rotemberg Weights

Instrument (Bartik Form): (type h , time t)

$$Z_{rt} \equiv \sum_h s_{rh,t-2} \Delta_{ht}, \quad (Z_{h,t})_{r\tau} = \begin{cases} s_{rh,t-2} \Delta_{ht}, & \tau = t \\ 0, & \tau \neq t \end{cases}, \quad g_{ht} \equiv \Delta_{ht}$$

Rotemberg Decomposition: (Goldsmith-Pinkham et al. (2020))

$$\hat{\beta}^{\text{Bartik}} = \sum_h \hat{\alpha}_h \hat{\beta}_h, \quad \hat{\alpha}_h \equiv \sum_t \hat{\alpha}_{h,t}, \quad \hat{\beta}_h \equiv \sum_t \left(\frac{\hat{\alpha}_{h,t}}{\hat{\alpha}_h} \right) \hat{\beta}_{h,t}$$

$$\hat{\beta}_{h,t} = (Z'_{h,t} X^\perp)^{-1} Z'_{h,t} Y^\perp \quad (\text{pseudo-2SLS})$$

$$\hat{\alpha}_{h,t} = \frac{g_{ht} Z'_{h,t} X^\perp}{\sum_{h',t'} g_{h't'} Z'_{h',t'} X^\perp}, \quad \sum_{h,t} \hat{\alpha}_{h,t} = 1$$